



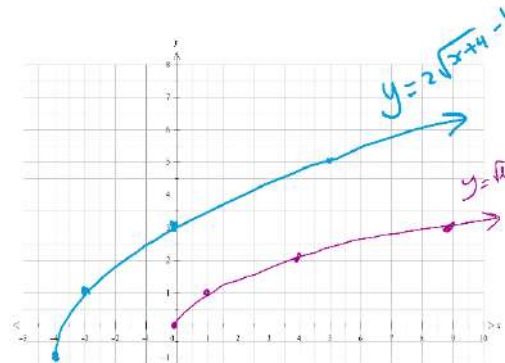
Transformation Topic Test 1

Solution

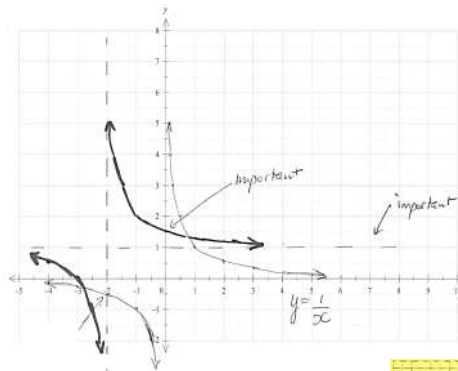
Total Marks: 30

Time: 50 minutes

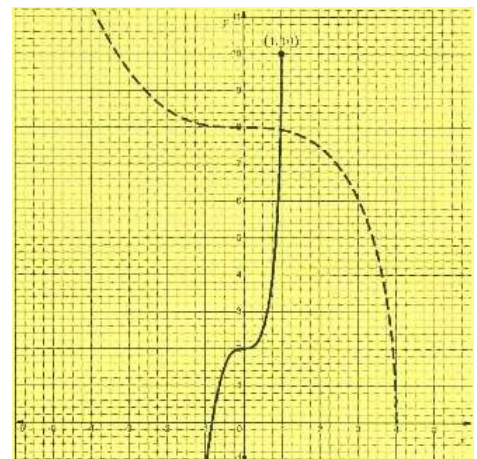
- 1 On the axes provided below, draw a sketch of $y = \sqrt{x}$ and use this to draw a sketch of $y = g(x) = 2\sqrt{x+4} - 1$ on the same set of axes. 3



- 2 On the same set of axes, draw a sketch of $y = \frac{1}{x}$ and use this to help you draw a sketch of $y = g(x) = \frac{1}{x+2} + 1$. 3



- 3 The diagram below shows the graph of $y = g(x)$. Sketch the curve of $y = 10 - g(4x)$ on the same set of axes. 3



- 4 Which of the following transformations do not commute? 1

- A. A vertical dilation and a horizontal dilation
- B. A vertical translation and a horizontal dilation
- C. A vertical translation and a vertical dilation
- D. A vertical translation and a horizontal translation

A translation and a dilation don't commute if both are vertical or horizontal.

5 Describe, in order, the transformations that are applied to $g(x) = e^x$ to form the function

3

$h(x) = 2e^{3x-1} + 1$. Stating the scale factors of any dilations involved.

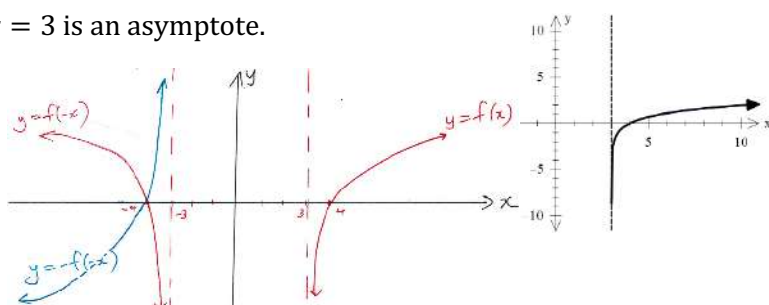
$$h(x) = 2e^{3(x-\frac{1}{3})} + 1$$

- Horizontal dilation of scale factor $\frac{1}{3}$
- Horizontal translation of $\frac{1}{3}$ unit to the right
- Vertical dilation of scale factor 2
- Vertical translation of 1 unit up

Note: there are many possible correct transformation, however this is the best one!

6 The graph of $f(x)$ is shown below, $x = 3$ is an asymptote.

(i) Sketch the graph of $-f(-x)$.



1

1

(ii) State the domain for $-f(-x)$. Give your answer in interval notation.

$$D: (-\infty, -3) \text{ (interval notation)}$$

7 (i) Express $y = 2x^2 - 12x + 23$ in the form $y = 2(x - c)^2 + d$.

2

$$\begin{aligned} y &= 2x^2 - 12x + 23 \\ y &= 2(x^2 - 6x) + 23 \\ y &= 2(x^2 - 6x + 9) + 23 - 18 \\ y &= 2(x-3)(x-3) + 5 \\ y &= 2(x-3)^2 + 5 \end{aligned}$$

2

(ii) The graph $y = x^2$ is transformed into the graph of $y = 2x^2 - 12x + 23$ by three transformations. The first transformation is a horizontal translation of 3 units to the right. State the other two translations in the correct order. Provide as much detail about the transformations as possible.

$$\begin{aligned} y &= x^2 \\ y &= (x-3)^2 && \text{HT 3 units right} \\ \frac{y}{2} &= (x-3)^2 \\ y &= 2(x-3)^2 && \text{VD SF 2} \\ &&& \text{vertical dilation with scale factor of 2} \\ &&& \text{vertical stretch along the y-axis} \\ &&& \text{or towards the x-axis} \\ y-5 &= 2(x-3)^2 \\ y &= 2(x-3)^2 + 5 && \text{VT 5 units up} \\ &&& \text{vertical translation of 5 units up.} \end{aligned}$$

- 8 The equation of the tangent to the curve $y = f(x)$ at the point $(2, 7)$ is $y = 5x - 3$. What is the equation of the tangent to the curve $f(x + 2) + 1$ at the point $(0, 8)$? 2

Q. Horizontally translated by 2 units to the left
 Q. Vertically translated by 1 unit up
 $(2, 7) \rightarrow (2-2, 7+1) = (0, 8)$
 $y = 5x - 3$
 New equation: $y = [5(x+2) - 3] + 1$
 $y = 5x + 8$

- 9 The graph of $y = \sqrt{x}$ undergoes the following sequence of transformations: 3

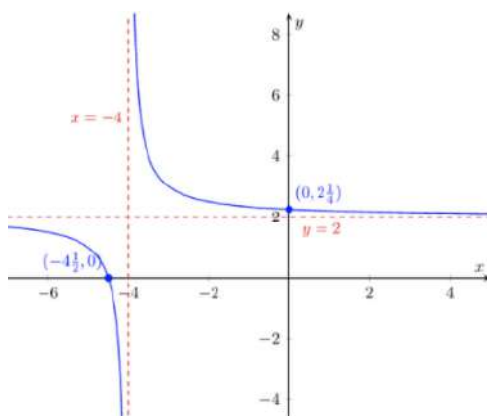
- Translation 3 units right
 - Dilation horizontally by a factor of $\frac{1}{2}$
 - Translation 1 unit down
 - Dilation vertically by a factor of 4
- Q. Translates 3 right: $x \rightarrow x-3: y = \sqrt{x-3}$
 Q. D. H. S.F. $\frac{1}{2}: x \rightarrow 2x: y = \sqrt{2x-3}$
 Q. Translates 1 down: $f(x)-1: y = \sqrt{2x-3}-1$
 Q. D. V. S.F. 4: $f(x) \rightarrow 4f(x): y = 4(\sqrt{2x-3}-1)$

Find the rule of the resulting graph.

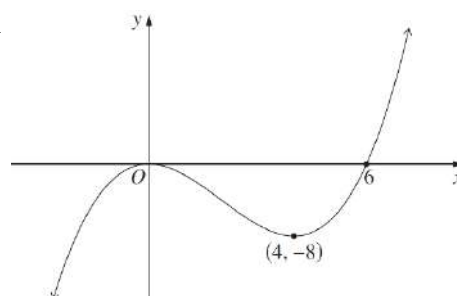
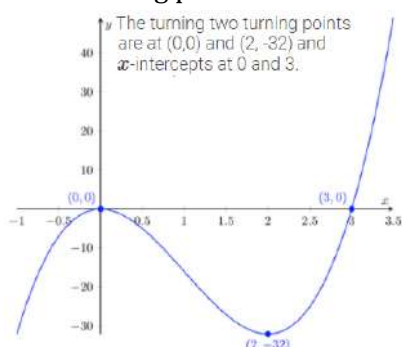
- 10 The function $f(x) = 4x^2$ is transformed by a horizontal translation 1 unit to the right followed by a horizontal dilation with scale factor 2. What is the equation of the new function $g(x)$? 2

$$f(x) = 4x^2, \quad x \rightarrow x-1 \rightarrow \frac{x}{2}-1 \quad \therefore g(x) = 4\left(\frac{x}{2}-1\right)^2 = (x-2)^2$$

- 11 Without using calculus, sketch the graph of $y = 2 + \frac{1}{x+4}$, showing the asymptotes and the x and y intercepts. 2



- 12 Consider the graph of $y = f(x)$ as shown. Sketch the graph of $y = 4f(2x)$ showing the x -intercepts and the coordinates of the turning points. 2





Transformation Topic Test 2

Solution

Total Marks: 30

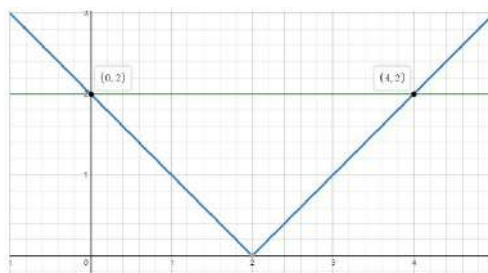
Time: 50 minutes

- 1 Write down the equation of the new curve when $f(x) = 3x + 2$ is translated left by 3 units. 2

- 2 (i) Solve the equation $|x - 2| = 2$. $x - 2 = \pm 2$ $x = 4$ or $x = 0$ 1

- (ii) Sketch $y = |x - 2|$ and $y = 2$ on the same number plane, clearly indicating the 2

coordinates of the points of intersection.

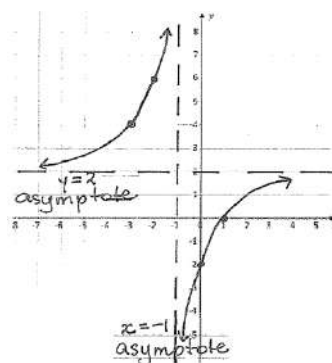


$$x \leq 0 \text{ or } x \geq 4$$

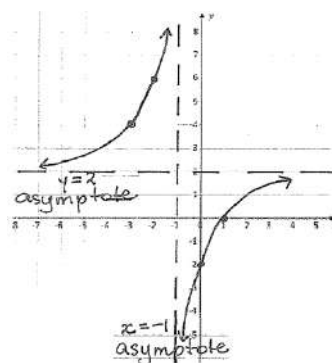
- (iii) Hence solve $|x - 2| \geq 2$. 1

- 3 The graph of the function $f(x) = 3x^{5/2}$ is reflected in the x - axis and then translated 3 units 2
to the right and 4 units down. What is the equation of the transformed function?

$$\begin{aligned} f(x) &= 3x^{5/2} \\ \text{after reflection: } y &= -3x^{5/2} \\ \text{after translation: } y &= -3(x-3)^{5/2} \\ \text{3 units right} \\ \text{after translation: } y &= -3(x-3)^{5/2} - 4 \\ \text{4 units down} \end{aligned}$$



- 4 Sketch the graph of $y = 2 - \frac{4}{x+1}$ on the axes below, showing 2
all intercepts with the coordinate axes and all asymptotes.



- 5 The graph $y = x^2 + 4x + 7$ is reflected in the y - axis, followed by a translation 2 units to the 2
right. What is the equation of the new graph?

$$\begin{aligned} \text{reflect in } y\text{-axis } y &= (-x)^2 + 4(-x) + 7 \\ &= x^2 - 4x + 7 \\ \text{translate 2 right } y &= (x-2)^2 - 4(x-2) + 7 \\ &= (-x+2)^2 + 4(-x+2) + 7 \end{aligned}$$

- 6 Let $y = \frac{3}{(x-3)(x+1)}$. 1

- (i) State the natural domain. $x \neq 3, -1$ 1

- (ii) Find the y - intercept. $y = -1$ 1

- (iii) Show that $y = 0$ is a horizontal asymptote. 1

By expanding the denominator then dividing all terms by x^2 and then allowing x to approach infinity. By the theory of limits $y = 0$ is the horizontal asymptote.

- (iv) Draw up a table of signs. 1

x	-1.01	-1	-0.99	0	2.99	3	3.01	4
y	74.81	*	-75.19	-1	-75.19	*	74.81	0.6
sign	+	*	-	-	-	*	+	+

- (v) Identify the vertical asymptote, and use the table of signs to write down its behaviour near them.

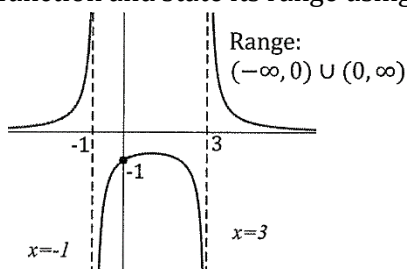
Vertical asymptotes: $x = 3$ and $x = -1$.

As $x \rightarrow 3^-$, $y \rightarrow -\infty$ As $x \rightarrow -1^-$, $y \rightarrow +\infty$

As $x \rightarrow 3^+$, $y \rightarrow +\infty$ As $x \rightarrow -1^+$, $y \rightarrow -\infty$

1

- (vi) Sketch the graph of the function and state its range using bracket interval notation.



1

- 7 Consider the circle given by the equation $x^2 + 8x + y^2 - 4y - 29 = 0$. Given the equation of the circle in the form $(x - h)^2 + (y - k)^2 = r^2$, if it is translated up by three units and right by five units.

$$x^2 + 8x + 16 + y^2 - 4y + 4 = 29 + 64 + 4$$

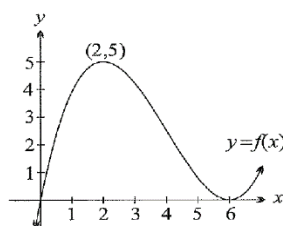
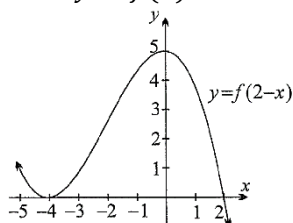
$$(x + 4)^2 + (y - 2)^2 = 49$$

Translate three units up and five units right

$$(x + 4 - 5)^2 + (y - 2 - 3)^2 = 49$$

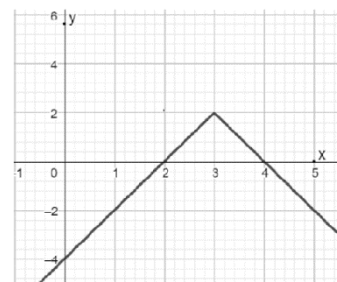
$$(x - 1)^2 + (y - 5)^2 = 7^2$$

- 8 The sketch shows $y = f(x)$. Draw a sketch of $f(2 - x)$.



2

- 9 The function $f(x) = |x|$ is transformed and the equation of the new function is of the form $y = kf(x + b) + c$, where k , b and c are constants. The graph of the new function is shown. What are the values of k , b and c ?



2

Translated to the right by 3 units
Reflected through the x -axis
multiplied by 2
Translated up by 2 units

$$\therefore k = -2$$

$$b = -3 \text{ and}$$

$$c = 2$$

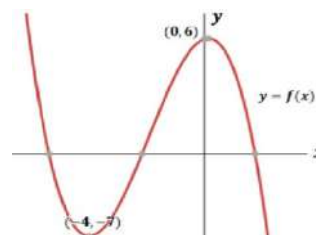
- 10 The graph of $y = f(x)$ has a stationary point at $(2, -3)$. What is the stationary point of

$$y = -f\left(\frac{x}{2}\right) - 5?$$

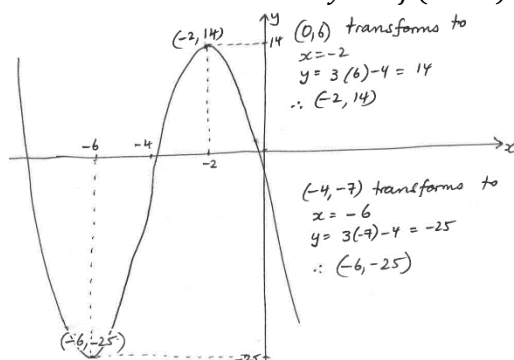
$$(4, -2)$$

1

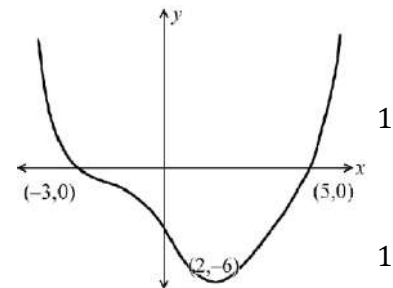
- 11 Given the graph of the function $y = f(x)$ below, with turning points as shown, sketch the transformed function $y = 3f(x + 2) - 4$.



2



- 12 The diagram shows a sketch of the graph of $y = f(x)$. The graph passes through the points $(-3, 0)$ and $(5, 0)$, and has a minimum turning point at $(2, -6)$.



- (i) The graph of $y = 4f(x + a)$ passes through the origin.

Write down the possible values of a . $a = -3, 5$

- (ii) The curve $y = f(x)$ is now transformed so that the

y - coordinate of the stationary point on the graph of $y = bf(x + 2)$ is 4. Write down the value of b .

$$\begin{aligned} bx - 6 &= 4 \\ b &= -\frac{2}{3} \end{aligned}$$